

PEICHI HUANG

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Education

University of Michigan, Ann Arbor

M.S, Data Science - GPA: 3.94/4.0

- Machine Learning, Optimization, Data Mining, Image Processing, Information Retrieval, Parallel Programming.

Aug. 2024 – Present

Michigan, United State

National Tsing Hua University

B.S, Mathematics and Computer Science - Major-GPA: 3.84/4.0 (Academic Excellence Award)

Sep. 2019 – Jan. 2024

Hsinchu City, Taiwan

Technical Skills

Programming Languages: Python, C, C++, Java, SQL, Julia, R, JavaScript, Kotlin

ML & Data: PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy, Spark, HuggingFace, LangChain, FAISS, CUDA

Systems & Tools: Git, Google Cloud Platform, AWS, Colab, Vue, React, REST APIs, Node.js, Linux

Experience

School of Information, University of Michigan

Graduate Student Instructor

Aug. 2025 – Dec. 2025

Michigan, US

- Assisted instruction for a graduate-level Information Retrieval course covering search, ranking, and embeddings.
- Mentored students on Python-based retrieval and ML projects through office hours and debugging support.

Blockchain Security Corporation

R&D Intern

Jun. 2025 – Aug. 2025

Taipei City, Taiwan

- Built an end-to-end OCR pipeline to identify cryptocurrency recovery seed phrases in real-world images, combining image preprocessing, PaddleOCR text recognition, and Trie-based matching over the BIP-39 dictionary.
- Designed a lightweight Laplacian blur-based filter to reduce unnecessary OCR calls and increase recall by 40%.

Information Technology Service Center, NYCU

Research Assistant

Feb. 2024 – Aug. 2025

Taipei City, Taiwan

- Implemented and evaluated a retrieval-based GPT-3.5 customer-service chatbot on the university website, tuning prompts and sampling temperature to reach 0.88 QA accuracy and 0.85 BERT F1 on 1,730 real user queries
- Performed EDA and feature engineering on post-graduate medical training outcome data, then trained regression and random forest models achieving >0.85 accuracy to provide actionable insights on resident performance and retention.

Biomedica Corporation

R&D AI Engineer Intern

Mar. 2024 – Aug. 2024

New Taipei City, Taiwan

- Trained deep learning models with YOLOv8 and YOLOv9 to automate medical image segmentation on disease-related datasets, achieving average mAP >0.90 and >0.95 with data augmentation.

Institute of Information Science, Academia Sinica

Software Engineer Intern

Jul. 2022 – Jan. 2024

Taipei City, Taiwan

- Engineered an Android-based digital diary application for GERD patients, enabling clinical data collection from 200+ patients and integrating an online questionnaire module for longitudinal health tracking.

Projects

MRI Denoising with Collaborative Non-Local Means | Python, Image Processing, Computer Vision

Winter 2026

- Reproduced and extended the MRI denoising method Collaborative Non-Local Means on the BrainWeb dataset.
- Accelerated the pipeline through vectorization and experimented with alternative weighting functions and adaptive patch/search window selection to improve PSNR up to 40.75 db and SSIM up to 0.97.

CUDA Optimization for CNN Convolution | C++, GPU Programming, CUDA

Fall 2025

- Optimized CUDA kernels for CNN convolutions using output parallelism, shared/constant memory, memory coalescing, and occupancy tuning, achieving $341\times$ speedup over serialized baseline (13.6 ms $\rightarrow$ 0.0398 ms).

SAIT, Study Abroad Information for Taiwanese Students | Vue, Git, Python, ML

Fall 2025

- Built a similarity-based admission analysis tool that recommends relevant study abroad cases to users based on profile embeddings and weighted similarity metrics.

GUI-ASFormer, Detecting Keyframes in GUI Videos | Python, ML, DL

Winter 2025

- Extended ASFormer for GUI action segmentation with high-resolution embeddings, frame differencing, and multi-scale temporal convolutions, improving F1@50 by +7.1 and edit distance by +6.8 on GUI video benchmarks.

Job Postings Searching System | Python, Information Retrieval, RAG

Fall 2024

- Implemented a retrieval-augmented job search system that uses user profiles as queries to match relevant job postings via text classification, semantic similarity, and ranking models, achieving 67% improvement in NDCG over baseline ranking.